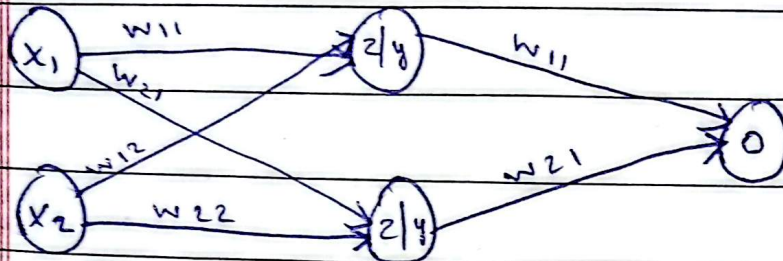


DATE: _____

DAY: _____

Name

Huzaifa Subhani

Input
layerHidden
layer

output

Solution

$$z_1 = x_1(w_{11})' + x_2(w_{12}) + b$$

$$z_1 = 0.5(0.1) + 0.1(0.2) + 0$$

$$z_1 = 0.05 + 0.02 + 0$$

$$z_1 = 0.07$$

$$z_2 = x_1(w_{21})' + x_2(w_{22}) + b$$

$$z_2 = 0.5(0.3) + 0.1(0.4) + 0$$

$$z_2 = 0.15 + 0.04 + 0$$

$$z_2 = 0.19$$

Sigmoid function

$$h_1 = \frac{1}{1 + e^{-0.07}}$$

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$$h1 = \frac{1}{1 + (2.71828)^{-0.07}}$$

$$h1 = 0.5175$$

$$h2 = \frac{1}{1 + (2.71828)^{-0.19}}$$

$$h2 = 0.5474$$

Output

$$\hat{y} = h1(w1) + h2(w2)$$

$$\hat{y} = 0.5175(0.5) + 0.5474(0.6)$$

$$\hat{y} = 0.587$$

Sigmoid

$$\hat{y} = \frac{1}{1 + (2.71828)^{-0.587}}$$

$$\hat{y} = 0.642676$$

Loss function

$$-1 \log(0.642676) - (1-1) \log(1-0.6426)$$

$$\text{Loss} = 0.443$$